

TECHNICAL BULLETIN

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**NADIC® METHYL ANHYDRIDE
(NMA)
CAS Reg. No. 25134-21-8**

Typical Properties

Properties	Spec/Target	Typical Value
Molecular Weight (average)	NA	178.2
Specific Gravity @ 20 °C, g/cc	1.200 - 1.250	1.226 – 1.233
Appearance	Pale yellow to tan liquid	Pass
Viscosity @ 25 °C, cP	300 max.	250
Refractive Index, n _{25/D}	1.500-1.506	1.505
Color, Pt-Co	75 max.	8 - 35
Anhydride Strength, %	99 min	99.4
NM Acid, %	1.0 max	0.1
Neutralization Equivalent, %	87 min	88 - 92
Solubility @ 25 °C	Acetone, benzene, naphtha, xylene	NA

Applications

NADIC methyl anhydride (NMA) is a liquid alicyclic anhydride commonly used to cure epoxy resins in many challenging applications, including fiber reinforced composites used in high performance aerospace and military applications, as well as mechanically demanding applications like filament wound bearings. The excellent electrical properties and high temperature performance make NMA-cured epoxies outstanding materials for encapsulating electronic components

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and circuits. NMA demonstrates many excellent characteristics when used as a curing agent for epoxy resins, including:

- Ease of handling
- Compatibility with solid and liquid epoxy resins
- Long pot life
- Low viscosity mixtures
- Low volatility
- Free of odor
- Less reactive than amines
- Easily catalyzed to match any process requirements
- Low exotherm

Epoxy resins cured with NMA demonstrate outstanding properties, including:

- Light color
- Excellent electrical characteristics, including arc resistance
- High heat distortion temperature and glass transition temperature
- Thermal stability
- Minimal shrinkage when cast
- Excellent hardness and impact strength
- Excellent chemical/solvent resistance

NMA may also be used as a building block for unsaturated polyester resins.

For more information on the use of anhydrides like NMA as epoxy curing agents, please consult the Technical Bulletin, "Formulating Anhydride-Cured Epoxy Systems," available from Dixie Chemical Company.

Anhydride Level

A range of 60-90 phr NMA can be used with standard BPA liquid epoxy resins. Most high performance applications utilize 80-90 phr where high levels of heat resistance, chemical resistance, or high strength are required.

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Examples

A study was done to examine the effect of anhydride level on performance. The levels of NADIC methyl anhydride chosen were 75, 85, and 93 phr to cure BPA liquid epoxy resin using 1 phr BDMA catalyst. The cure cycle was 2 hours at 93°C with a post-cure of 2 hours at 204°C.

The following results were obtained:

NMA, phr	75	85	93
HDT, °C	129	132	102
Tensile Strength, psi	11,000	11,000	11,600
Elongation, %	3.0	2.7	2.8
Flexural Strength, psi	18,900	20,700	24,500
Flexural Modulus, ksi	520	570	560
Compressive Strength, psi	18,700	19,400	20,500
Dielectric Constant @ 25°C 1 MHz	3.6	3.8	3.7
Dissipation Factor @ 25°C 1MHz	0.01	0.02	0.02

These results display a typical response for heat distortion temperature (HDT), and glass transition temperature (T_g) where there is an optimum anhydride level for a given formulation (epoxy and catalyst) and cure cycle.

Cure Catalysts

Since the reactivity of NMA is slightly lower than Dixie's other ECA products, latent quaternary ammonia salts might be too slow for the curing. It usually requires active catalysts such as tertiary amines or Imidazoles.

Cure Cycle

The cure cycle used has a dramatic effect on system performance. This cure cycle includes the initial time and temperature used to cure the formulation, as well as all post-cure times and temperatures used to improve performance properties. This is important because longer cure times and higher cure

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temperatures promote increases in crosslinking. Higher crosslinking results in higher levels of mechanical strength and chemical resistance.

Formulating with Liquid Epoxy Resins

Dow Chemical Company conducted an excellent and extensive formulation study in which a variety of liquid epoxy resins were formulated with NADIC methyl anhydride.¹ It is worthwhile to review some of these results here. The table below summarizes key formulation and performance data from this study. It is noted that all formulations gave high levels of hardness and strength, combined with good toughness and elongation. All formulations demonstrated excellent electrical properties.

¹ "Dow Liquid Epoxy Resins," published by Dow Chemical Company, January, 1999

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Properties of Liquid Epoxy Resins with Nadic Methyl Anhydride

Dow Liquid Epoxy Resin ¹	Condition	D.E.R. 332	D.E.R. 330	D.E.R. 383	D.E.R. 331	D.E.R. 317
Average Epoxy Equivalent Weight		147	180	180	190	197
Average Epoxy Viscosity, P		50	85	97.5	130	200
Nadic Methyl Anhydride, phr		87.5	87.5	87.5	87.5	87.5
Formulation Viscosity @ 80°C, cP		30	35	36	38	45
Gel Time, min ²		97	158	160	129	80
Gel Temperature, °C ²		93	113	112	99	100
Peak Exotherm Time, min ²		152	132	130	146	155
Peak Exotherm, °C ²		125	180	182	153	100
Heat Distortion Temperature, °C ³		135	148	144	156	147
Flexural Strength, psi ⁴		21,200	19,200	18,500	14,000	15,000
Flexural Modulus, ksi ⁴		472	470	480	440	441
Compressive Strength, psi ⁵		20,190	16,900	17,100	18,300	15,000
Compressive Modulus, ksi ⁵		340	384	380	440	441
Tensile Strength, psi ⁶		6,260	6,340	7,000	10,000	7,000
Elongation, % ⁶		1.6	1.4	1.7	2.5	1.8
Izod Impact, ft-lb/in. notch ⁷		0.21	0.30	0.3	0.48	0.48
Hardness, Rockwell M ⁸		114	111	112	111	109
Dielectric Constant (Condition A) ⁹	@ 60 Hz	3.14	3.15	-	3.15	3.12
	@ 1 KHz	3.12	3.13	3.54	3.14	3.09
	@ 1 MHz	2.99	3.01	-	2.97	2.89
Dielectric Constant (Condition D) ⁹	@ 60 Hz	3.30	3.39	-	3.34	3.22
	@ 1 KHz	3.28	3.35	-	3.32	3.19
	@ 1 MHz	3.13	3.14	-	3.11	3.01
Dissipation Factor (Condition A) ⁹	@ 60 Hz	0.0049	0.0030	-	0.0020	0.0024
	@ 1 KHz	0.0045	0.0054	0.0038	0.0054	0.0053
	@ 1 MHz	0.015	0.016	-	0.017	0.018
Dissipation Factor (Condition D) ⁹	@ 60 Hz	0.0030	0.0079	-	0.0023	0.0038
	@ 1 KHz	0.004	0.0063	-	0.0059	0.0059
	@ 1 MHz	0.018	0.020	-	0.21	0.020
Volume Resistivity, ohm-cm ¹⁰	Condition A	7.20x10 ¹⁵	6.11x10 ¹⁵	6.15x10 ¹⁵	6.10x10 ¹⁵	0.90x10 ¹⁵
	Condition D	1.01x10 ¹⁵	3.67x10 ¹⁵	-	1.17x10 ¹⁵	0.47x10 ¹⁵
Surface Resistivity, ohm-cm ¹⁰	Condition A	6.28x10 ¹⁵	4.71x10 ¹⁵	4.95x10 ¹⁵	3.93x10 ¹⁵	>7.85x10 ¹⁵
	Condition D	1.1x10 ¹⁵	3.93x10 ¹⁵	-	3.93x10 ¹⁵	3.93x10 ¹⁵

¹ Data adapted from "Dow Liquid Epoxy resins" published by Dow Chemical Company, January, 1999

² 500g samples held @ 80 °C

³ ASTM D648

⁴ ASTM D790

⁵ ASTM D695

⁶ ASTM D638

⁷ ASTM D256

⁸ ASTM D785

⁹ ASTM D150

¹⁰ ASTM D257

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Formulating with Epoxy Bisphenol A Novolac Resin

Momentive Specialty Chemicals has an epoxy bisphenol A Novolac Resin, with the trade name EPON™ Resin SU-3. It is another multifunctional epoxy resin which gives very high performance in fiber reinforced composites. Momentive has published a formulation for a resin system based on this bisphenol A novolac and NADIC methyl anhydride.² This system is designed to be a “High Temperature FRP Matrix for Wet Lay-Up, Filament Winding and Prepreg.” It is suitable for electrical applications, aerospace applications, and other high performance composite applications. It is useful to review some features of this formulation here. However, one should also consult the original reference for more process details on laminating, filament winding, and prepreg fabrication, and for laminate performance characteristics.

The formulation consists of 100 parts EPON Resin SU-3, 88 parts NMA, and 0.5 parts 2-ethyl-4-methyl imidazole. Formulation viscosity and working life as a function of temperature are summarized as follows:

Temperature, °C	Initial Viscosity, cP	Working Life, h
25	8500	>24
49	1020	12
71	200	8
82	300	2

² "Starting Formulation 8011," published by Momentive Specialty Chemicals, July, 2007.

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The cure cycle is 2 hours at 93°C, 6 hours at 175°C, and 8 hours at 260°C. Performance characteristics include:

Properties	Conditions	Value
Heat Distortion Temperature, °C		276
Tensile Strength, psi		8500
Tensile Modulus, ksi		570
Elongation, %		1.5
Flexural Strength, psi	@ 25°C	10,700
	@ 175°C	7000
	@ 230°C	4300
Flexural Modulus, ksi	@ 25°C	510
	@ 175°C	330
	@ 230°C	230
Weight Loss on Heat Aging, %	1000 h @ 150°C	nil
	350 h @ 200°C	0.9
	700 h @ 200°C	1.6
	350 h @ 260°C	10.5
	700 h @ 260°C	16.3

The high temperature performance is clearly excellent.

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Handling & Storage

It is recommended that NADIC methyl anhydride is stored between the temperatures of 18° C (64° F) and 40° C (104° F) to avoid freezing, although the material is not affected by freezing. If material does freeze, thaw at a temperature below 40° C and mix well before using. Store in a dry place and keep container tightly closed to prevent absorption of atmospheric moisture or contamination. The presence of moisture could cause free acid to form in the anhydride.

NMA will react with water to form diacids. This is normally undesirable, so NMA should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for NMA, consult the Dixie Chemical brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

NMA is available from Dixie Chemical in Pasadena, Texas. Contact your Dixie representative for details.